

Bash, Sed, and Awk - Detailed Revision

Bash - Basics

- Shebang: `#!/bin/bash`
- Run script: `bash script.sh` or `./script.sh` (with `chmod +x`)
- Variables: `x=5` (no spaces). Access: `$x` or `${x}`
- Data types are all strings. Use `(( ))`, `let`, or `expr` for math.
- Input: `read -p "Enter: " var`

Bash - Control Flow and More

- Conditionals: `if [ $x -gt 5 ]; then ... elif ... else ... fi`
- Loops: `for`, `while`, `until`
- Functions: `greet() { echo Hello $1; }`
- Arrays: `arr=(1 2 3)`, `${arr[0]}`, `${arr[@]}`
- Strings: `${#x}`, `${x:1:3}`, `${x/foo/bar}`
- File tests: `-f`, `-d`, `-s`, `-r`, etc.
- Command Substitution: `now=$(date)`
- Redirection: `>`, `>>`, `<`, `|` etc.

Sed - Basics and Use

- Replace: `sed 's/foo/bar/' file`
- Global replace: `sed 's/foo/bar/g'`
- Case-insensitive: `sed 's/foo/bar/gi'`
- Delete lines: `sed '/pattern/d'`
- Print matching: `sed -n '/pattern/p'`
- In-place: `sed -i 's/foo/bar/g' file`
- Multiple commands: `sed -e 'cmd1' -e 'cmd2'`

Sed - Advanced

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- Address ranges: `sed '2,5s/foo/bar/'`
- Regex: `sed '/[0-9]\{3\}/d'`
- Insert/append from file: `sed '/pattern/r file'`

Awk - Structure and Fields

- Format: `awk 'BEGIN{} {block} END{}' file`
- Fields: `$1, $2, ..., $NF`. `$0` is full line.
- FS/OFS: `awk 'BEGIN{FS=":"; OFS=" - "}' ...`
- Print fields: `awk '{print $1, $3}'`

Awk - Logic and Functions

- Conditions: `awk '$2 > 50 {print $1}'`
- String functions: `length()`, `substr()`, `tolower()`, `toupper()`
- Built-ins: `NR`, `NF`, `FNR`, `RS`, `ORS`
- User variables: `awk -v var=val '{print var}' file`
- Multi-file: `awk '{print FILENAME, $0}' file1 file2`